



Internal heating of lithium-ion batteries using alternating current based on the heat generation model in frequency domain



Jianbo Zhang, Hao Ge, Zhe Li*, Zhanming Ding

Department of Automotive Engineering, State Key Laboratory of Automotive Safety and Energy, Tsinghua University, 100084 Beijing, China

HIGHLIGHTS

- An internal AC heating method is developed for cells at low temperatures.
- A heat generation rate model in frequency domain is developed to simulate the cell temperature rise.
- The AC amplitude and frequency have significant impact on the heat generation rate.

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ABSTRACT

This study develops a method to internally preheat lithium-ion batteries at low temperatures with sinusoidal alternating current (AC). A heat generation rate model in frequency domain is developed based on the equivalent electrical circuit. Using this model as the source term, a lumped energy conservation model is adopted to predict the temperature rise. These models are validated against the experimental results of preheating an 18650 cell at different thermal insulation conditions. The effects of current amplitude and frequency on the heating rate are illustrated with a series of simulated contours of heating time. These contours indicate that the heating rate increases with higher amplitude, lower frequency and better thermal insulation. The cell subjected to an alternating current with an amplitude of 7 A (2.25 C) and a frequency of 1 Hz, under a calibrated heat transfer coefficient of $15.9 \text{ W m}^{-2} \text{ K}^{-1}$, can be heated from -20°C to 5°C within 15 min and the temperature distribution remains essentially uniform. No capacity loss is found after repeated AC preheating tests, indicating this method incurs little damage to the battery health. These models are computationally-efficient and can be used in real time to control the preheating devices in electric vehicles.

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1. Introduction

The resistance of lithium-ion batteries drastically increases at low temperatures, causing the loss of usable energy even at normal discharge rate. More severely, the batteries can hardly be charged at subzero temperature. If charging is forcibly performed under this circumstance, lithium is inclined to deposit on the surface of negative electrodes in the form of dendrite that may reach the positive electrode and cause internal short, inducing safety hazards such as thermal runaway. Generally, lithium-ion batteries need to be preheated in cold weather before charge till the batteries reach a charging-friendly temperature like above 0°C or even 5°C .

The batteries can be heated either externally or internally [1]. External heating always adopts electrical heating actuators. These

actuators are embedded in a blanket wrapped outside the battery pack, or immersed in the flow channels using liquid or air as the medium to heat the battery [1,2]. External heating can be slow and inefficient due to the loss of energy to the environment in each intermediate step of the heat conduction. Besides, it could easily cause temperature non-uniformity in the battery pack, accelerating the aging at local hot spots [1,3]. In contrast, the internal method heats the battery with the heat generated inside the battery itself during charging and/or discharging, eliminating the long paths of heat conduction and avoiding the heat accumulation in the vicinity of heating devices. Therefore, internal heat method has the potential to provide fast heating with higher efficiency and less thermal-induced aging stress.

Both direct current and alternating current (AC) can generate heat inside the battery. When a direct current is used to heat the battery, both the amplitude and the duration of the current should be restricted to very low values to avoid lithium deposition, thereby limiting the heat generation rate and the preheating effectiveness.

* Corresponding author.

E-mail address: lizhe1212@gmail.com (Z. Li).

In contrast, alternating current can heat the battery utilizing the real part of the battery impedance, while avoiding substantial change of the SOC (state-of-charge) and the lithium deposition. AC heating can be applied for an extended period of time till the heat generation rate approaches the heat removal rate which increases with the rise of the battery temperature.

Pesaran et al. [3] compared several preheating methods. They commented that compared with external heating, internal heating achieved not only a faster heating rate but also more uniform temperature distribution. Furthermore, AC heating could heat up the battery with less damage (e.g. less capacity loss) than DC heating. Hande et al. [1,4,5] developed the internal preheating method using alternating current for NiMH batteries at low temperatures, and designed a high-frequency inverter to generate the alternating square-wave current. However, these studies have not discussed how to choose the parameters in the AC heating. Yan et al. [6] developed a thermal–electrochemical model to simulate the AC heating process and got satisfactory results. However, this thermal–electrochemical model is in the time domain. It cannot directly show the effects of current frequency on the heat generation rate. Moreover, modeling high frequency AC heating process using such a complicated model represents a great computational challenge, limiting its usage in real preheating devices in the battery management systems (BMS).

This study developed a method to internally preheat lithium-ion batteries at low temperature with a sinusoidal alternating current. A model for the heat generation rate based on the battery equivalent electrical circuit (EEC) was developed in the frequency domain. Meanwhile, a lumped energy conservation model was adopted to predict the temperature rise with the heat generation model as the source item. The effects of preheating a cylindrical 18650 cell with different alternating current frequencies and amplitudes were both observed in tests and simulated using this model. The simulation results were then validated against experimental data in different thermal insulation conditions. The cell subjected to an alternating current with mild C-rates (2–3 C) amplitude and 1 Hz frequency could be warmed up from -20°C to $+5^{\circ}\text{C}$ within a few minutes and the temperature distribution remains essentially uniform. No capacity loss is found after repeated AC preheating tests, indicating that this method incurs little damage to the battery health. The developed models are computationally-efficient and can be used in real preheating devices of electric vehicles to determine the proper heating algorithm for batteries in real time.

2. Model development

In the long run, a battery subjected to an alternating current will be heated only by the irreversible heat due to the Joule effect, since the reversible heat associated with the electrochemical reaction is canceled within one period. For the discussion of heat generation

with alternating current, a description of the battery in the frequency domain, the equivalent electrical circuit model, will be convenient. The equivalent electrical circuit model used in this study is shown in Fig. 1.

The element Q_{SEI} and Q_{dl} in Fig. 1 stand for the constant phase element (CPE), and the element W stands for Warburg impedance [7]. CPE consists of a generalized capacitor Q and a depression factor n . The impedance of a CPE and a Warburg element is defined as:

$$\begin{aligned} Z_{\text{CPE}} &= \frac{1}{(j\omega)^n Q} = \frac{1}{\left(\cos \frac{n\pi}{2} + j \sin \frac{n\pi}{2}\right) \omega^n Q}, \\ Z_{\text{W}} &= \frac{1}{(j\omega)^{\frac{1}{2}} Q} = \frac{1}{\left(\cos \frac{\pi}{4} + j \sin \frac{\pi}{4}\right) \omega^{\frac{1}{2}} Q} \end{aligned} \quad (1)$$

respectively, where ω is the angular frequency of the current, and j is the imaginary unit which has: $j = e^{(\pi/2)j}$ and $j^n = e^{(n\pi/2)j} = \cos(n\pi/2) + j \sin(n\pi/2)$.

The equivalent electrical circuit is composed of four parts: Z_L , the inductance caused by metallic elements in batteries and connecting wires; Z_1 , the ohmic resistance; Z_2 , the impedance for the solid electrolyte interphase (SEI); and Z_3 , the impedance for the charging and charge transfer processes at the electrolyte–electrode interface, and the transport of active species within or outside of the active materials [8]. The effect of the inductance L is only obvious at very high frequency and has no contribution to the real part of the impedance, so Z_L is neglected. The Warburg impedance, related to the diffusion process, is only obvious at very low frequency region. Moreover, the characteristic frequency of the diffusion process shifts to a lower value when the temperature is low. Thus, the Warburg diffusion impedance is quite small in the frequency region ($f > 0.1$ Hz) used in this study. Therefore, the Warburg impedance is also neglected in this study. All the parameters in this circuit are temperature-dependent. In the frequency domain, the real part of the circuit impedance can be expressed as:

$$\begin{aligned} Z_{\text{Re}} &= R_{\Omega} + \frac{\frac{1}{R_{\text{SEI}}} + Q_{\text{SEI}} \omega^{n_{\text{SEI}}} \cos \frac{n_{\text{SEI}}\pi}{2}}{\left(\frac{1}{R_{\text{SEI}}}\right)^2 + \frac{2}{R_{\text{SEI}}} Q_{\text{SEI}} \omega^{n_{\text{SEI}}} \cos \frac{n_{\text{SEI}}\pi}{2} + (Q_{\text{SEI}} \omega^{n_{\text{SEI}}})^2} \\ &\quad + \frac{\frac{1}{R_{\text{ct}}} + Q_{\text{dl}} \omega^{n_{\text{dl}}} \cos \frac{n_{\text{dl}}\pi}{2}}{\left(\frac{1}{R_{\text{ct}}}\right)^2 + \frac{2}{R_{\text{ct}}} Q_{\text{dl}} \omega^{n_{\text{dl}}} \cos \frac{n_{\text{dl}}\pi}{2} + (Q_{\text{dl}} \omega^{n_{\text{dl}}})^2} \end{aligned} \quad (2)$$

where Q_{SEI} and n_{SEI} are the generalized capacitor and depression factor of CPE Q_{SEI} , Q_{dl} and n_{dl} are the generalized capacitor and depression factor of CPE Q_{dl} , R_{Ω} , R_{SEI} and R_{ct} are the resistances of the elements in Fig. 1.

When the applied current is $i(t) = A \sin(\omega t + \phi)$, the heat generation rate $\dot{Q}$ is

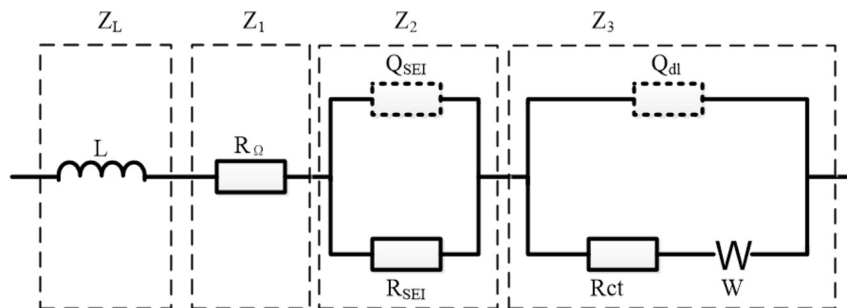


Fig. 1. Equivalent electrical circuit model.

$$\dot{Q} = \left(\frac{A}{\sqrt{2}} \right)^2 Z_{\text{Re}} \quad (3)$$

where $\dot{Q}$ is the heat generation rate, A is the amplitude of the alternating current, and Z_{Re} is the real part of the impedance of the equivalent electrical circuit.

Equation (2) shows that the impedance decreases with the increase of the current frequency. Thus, given a specific battery equivalent electrical circuit, the heat generation rate is determined both by the current amplitude A and the frequency ω .

In addition to the heat generation rate equation in the frequency domain, an energy conservation equation describing the relation between the rate of the temperature rise, the heat generation, and the thermal insulation conditions is also needed to simulate the temperature rise during internal AC preheating.

For 18650 cells, the internal and surface temperatures are quite close during discharge according to Ref. [9]. Thus, the battery temperature increase can be described by a lumped energy conservation equation:

$$mc \frac{dT}{dt} = \dot{Q} - hS(T - T_{\infty}) \quad (4)$$

where T is the battery temperature, m is the battery mass, c is the specific heat capacity, S is the surface area of the battery, h is the equivalent heat transfer coefficient, and T_{∞} is the ambient temperature. The equivalent heat transfer coefficient characterizes the heat transfer/exchange effect between the battery and the environment.

Equations (3) and (4) constitute the model for the prediction of the heat generation rate and the temperature rise of a battery under AC heating in this study. Due to the complex correlations between the values of the equivalent circuit elements and the temperature, numerical solutions are obtained with MATLAB programming.

3. Experimental

The battery tested in this study was a commercial 18650 cell and the specifications are shown in Table 1. The experiment devices and analytical software used in this study are also listed in Table 1. The experimental details are shown in Table 2.

Table 1
Specifications of the experimental devices, analytical software and test battery.

Item	Specification	Value
Experimental devices	EIS analyzer and the preheating load	Autolab PGSTAT302N (Eco Chemie, Netherlands)
	Environmental chamber	GDJW-225 (Yashilin, China)
	Data acquisition system	National Instrument PXIe-4353 system (National Instruments, America)
Analytical software	EEC model parameters fitting software	ZSimpWin (Princeton Applied Research, America)
Battery	Type	Cylindrical 18650 lithium-ion battery
	Chemical compositions	Cathode: $\text{Li}_x\text{Ni}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ Anode: graphite Electrolyte: LiPF_6 in organic solvent
	Capacity	2.8 A h
	Adjusted SOC	50%

Table 2
Experimental details.

Test	Item	Value
EIS tests	Mode	Potentiostatic
	Amplitude	5 mV
	Frequency range	10^4 to 10^{-1} Hz
	Data points	51 (10 points per decade)
	Temperature	-20°C , -18°C , -16°C , -14°C , -12°C , -8°C , -4°C , 0°C , 3°C , 5°C
Equivalent heat transfer coefficients calibration	Battery initial temperature	30°C
	Thermal insulation conditions	Condition 1, condition 2
	Ambient temperature	-20°C
AC preheating tests	Battery initial and ambient temperature	-20°C
	Thermal insulation conditions	Condition 1, condition 2
	Current parameters	3 A 1 Hz, 5 A 0.1 Hz,
		5 A 1 Hz, 5 A 10 Hz, 7 A 1 Hz

3.1. Measurement of the electrochemical impedance spectroscopy (EIS)

Equations (2) and (3) show that the heat generation rate depends on the parameters of the equivalent electrical circuit in Fig. 1. These parameters are fitted using the EIS data. The EIS data was measured using an Autolab PGSTAT302N impedance analyzer (Eco Chemie, Netherlands) in an environmental chamber GDJW-225 (Yashilin, China) and the EEC models were fit through the EIS data using the ZSimpWin software (Princeton Applied Research, America) as shown in Table 1. The EIS data was measured under a small sinusoidal excitation of 5 mV amplitude over the frequency range of 10^4 to 10^{-1} Hz at ten temperatures as shown in Table 2. The battery was soaked at each temperature in the environment chamber for more than 3 h to ensure temperature equilibration.

3.2. Calibration of the equivalent heat transfer coefficients

Equation (4) shows that the equivalent heat transfer coefficient h (i.e. the thermal insulation condition) influences the temperature rising rate. The effects of thermal insulation conditions on the temperature rise rate were investigated by adjusting the plastic box housing the battery. In one configuration, the battery is just placed inside one box. In the second configuration, the battery is placed inside a small box, which is nested inside a larger box. In this way, the heat transfer coefficient can be shifted and controlled reproducibly. The two conditions are shown in Fig. 2.

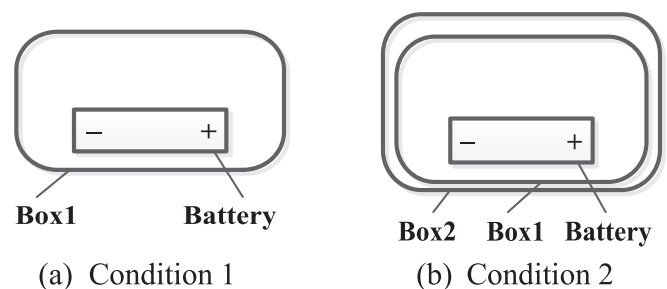


Fig. 2. Diagram of the two conditions.

The equivalent heat transfer coefficients were calibrated from the cooling curve of the battery heated to a high temperature. The battery was thermally equilibrated in one environmental chamber to 30 °C and then was put into the boxes, and shifted to another environmental chamber which was held at −20 °C. The battery temperatures were measured during the cooling process using two T-type thermocouples connected to a National Instrument system as shown in Table 1, one of which was affixed at a one-third position along the axis direction near the negative end and the other thermocouple at the other one-third position near the negative end. The equivalent heat transfer coefficients were calculated from the temperature curves, and the detailed calculation method is described in Section 4.2.

3.3. Preheating tests

The test battery was preheated from −20 °C using sinusoidal alternating current with various sets of frequencies and amplitudes for the two thermal insulation conditions as shown in Table 2. The battery was kept in the boxes in the chamber at −20 °C for more than 3 h before preheating. The current was generated by the Autolab PGSTAT302N as shown in Table 1 and the wires passed through small holes in the boxes. The heating process ended when the battery temperature reached 5 °C or the heating time was beyond half an hour. The battery temperatures were measured using six T-type thermocouples connected to a National Instrument system as shown in Table 1, with three thermocouples at a one-third position along the axis direction near the negative end and others at the other one-third position neat the positive end. Each three were evenly spaced in the circumferential direction to examine the uniformity of the temperature distribution. The test system is shown in Fig. 3.

4. Results and discussions

4.1. EIS results and the parameters of the EEC model

Fig. 4 shows the EIS results over the frequency range of 10^4 to 10^{-1} Hz at ten temperatures ranging from −20 °C to 5 °C. The results show that the EIS of this cell shrinks in both the real and the imaginary parts with the increase of the temperature. Besides, the EIS results at low temperatures show some structural changes as compared with the ones at higher temperatures. Since the characteristic frequency of the diffusion process shifts to lower values

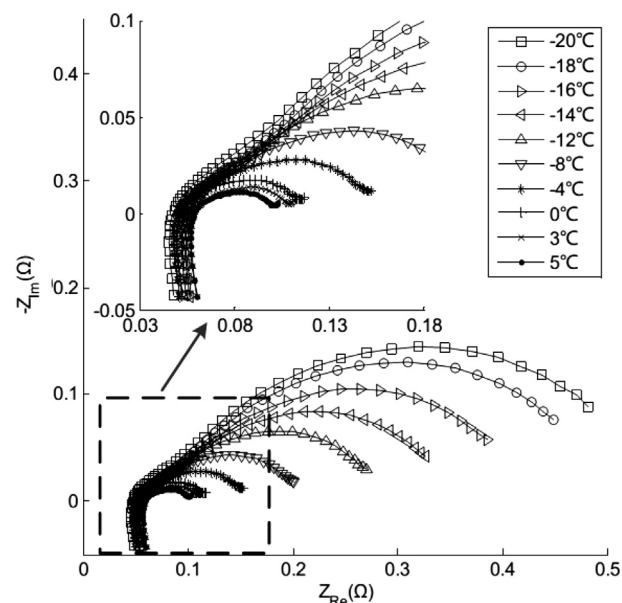


Fig. 4. EIS results at different temperatures.

outside the preset frequency range, the oblique line corresponding to the diffusion process finally vanishes in the measured spectrum at the low temperatures.

The impedance increases drastically with the decrease of temperature, implying that the parameters of the equivalent electrical circuit are highly temperature-sensitive. The equivalent electrical circuit model parameters at different temperatures are fitted using the EIS data. The results are shown in Fig. 5(a)–(c). Fig. 5(d) shows that the logarithm of charge transfer resistance (R_{ct}) varies linearly with the inverse of the absolute temperature [10], indicating an Arrhenius behavior with an activation energy of 74.7 kJ mol^{−1}. The charge transfer resistance, which reflects the electrochemical reaction rate, increases with the decrease of the temperature, becoming the dominant part of the impedance at temperatures below −5 °C.

4.2. Calibration of the equivalent heat transfer coefficients

During the cooling-down of the battery, the heat generation rate of the battery is zero and the equivalent heat transfer coefficients can be calculated from Eq. (4). The parameters used in the calculation are listed below in Table 3. The value of specific heat capacity is from Ref. [11]. The battery surface area S is calculated through the geometry of the cylindrical 18650 cell with the diameter of 18 mm and the length of 65 mm, while the other parameters are measured.

The temperatures measured by the two thermocouples are very close and the battery temperature is defined as the average of the two thermocouples. The equivalent heat transfer coefficient h is then fitted using the temperature data at the range of 5 °C to −15 °C.

The cooling curves in different thermal insulation conditions are displayed in Fig. 6. The equivalent heat transfer coefficient of condition 1 is 21.6 W m^{−2} K^{−1}, while the coefficient of condition 2 is 15.9 W m^{−2} K^{−1}.

4.3. Model validation and preheating performance

The maximum temperature variation among the six thermocouples is less than 2 °C, whereas the battery temperature rise is

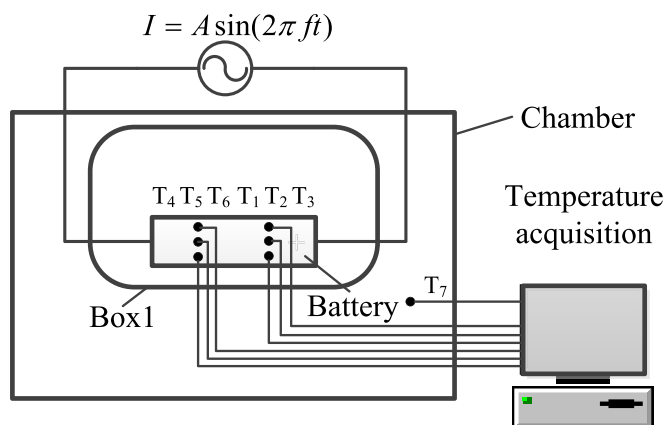


Fig. 3. Diagram of preheating test system.

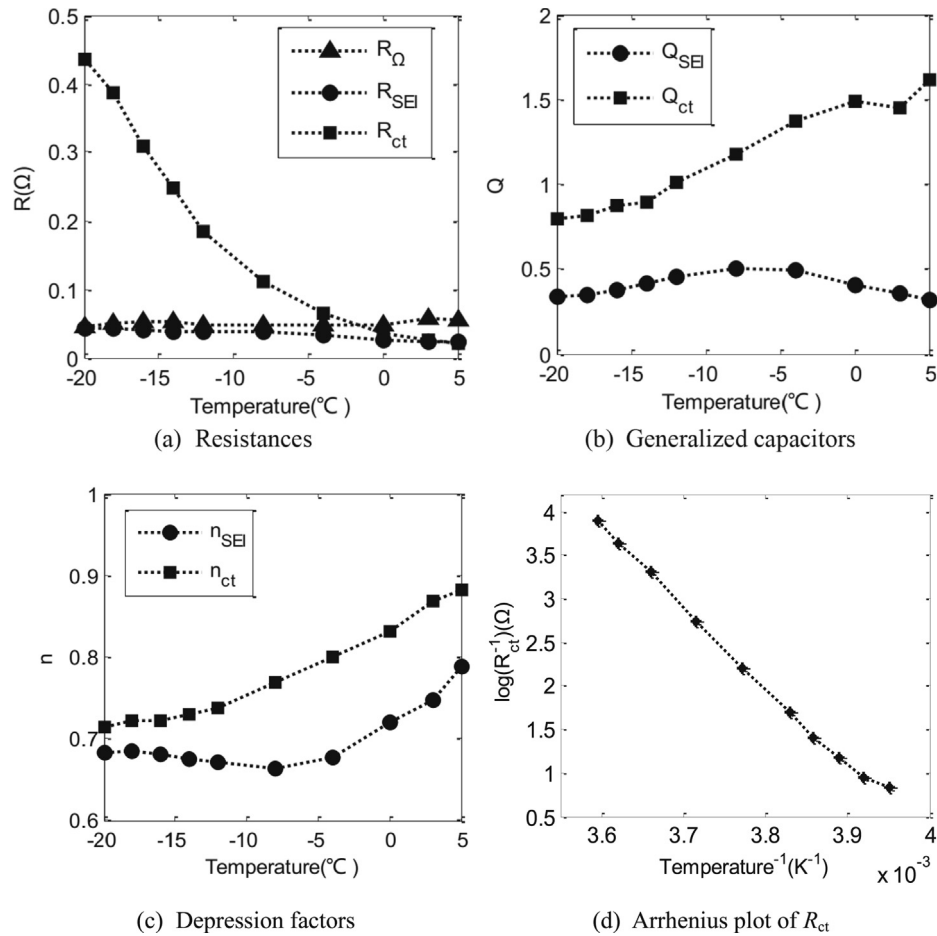


Fig. 5. Equivalent electrical circuit parameters at different temperatures.

about 20 °C. This reinforces the assumption of the lumped thermal model in Eq. (4).

The experimental and simulated temperature curves are shown in Fig. 7. The experimental temperature is defined as the average of the six thermocouples.

Fig. 7 shows good agreement between model predictions and experimental data. The maximum deviation, appearing at 5 A and 10 Hz in condition 2, is less than 1.5 °C. The results show that the developed models can predict the temperature rise of the 18650 battery with high precision during the preheating process using sinusoidal alternating current under different thermal insulation conditions. Moreover, with no need to solve the complicated thermal–electrochemical equations, the proposed models in this study directly calculate the heat generation rate from the real part of the impedance, improving its applicability in the BMS of electric vehicles due to its computationally efficiency.

Table 3
Parameters used in the calculation of h .

Parameter	Symbol	Value
Battery mass	m	46.0 g
Battery surface area	S	$4.18 \times 10^{-3} \text{ m}^2$
Specific heat capacity	c	$1.72 \text{ J g}^{-1} \text{ K}^{-1}$ [11]
Heat generation rate	$\dot{Q}$	0 W
Ambient temperature	T_{∞}	−20 °C

To further investigate the effects of current amplitude and frequency as well as the thermal insulation conditions on the heating rate, the needed preheating time to heat the battery from −20 °C to 5 °C was simulated using the developed model and the contours are shown in Fig. 8. Note that the scale for the frequency axis is logarithmic.

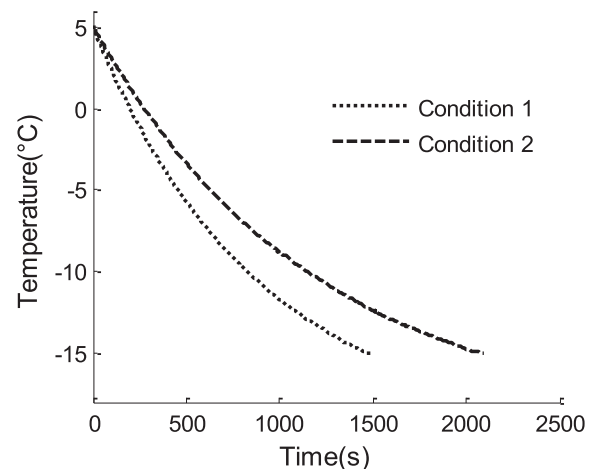
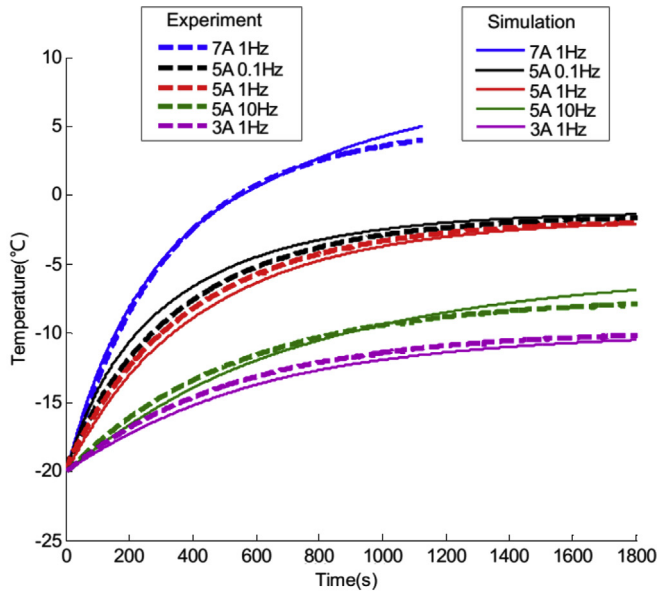
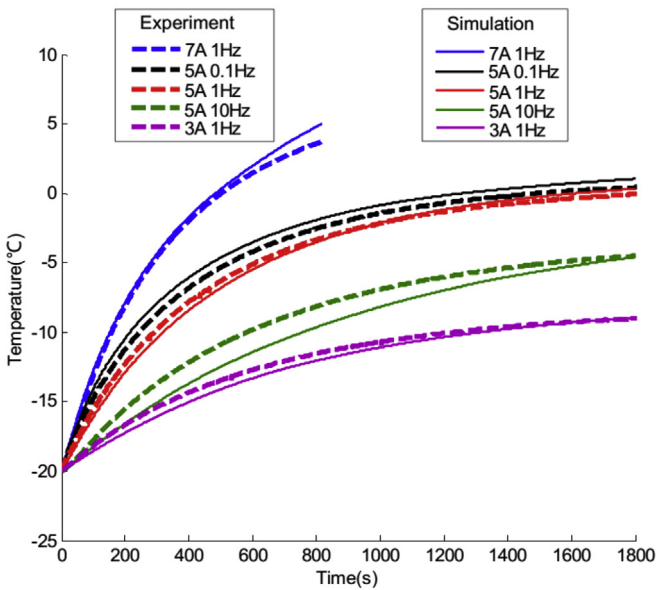


Fig. 6. Temperature cooling curves in different thermal insulation conditions.

(a) Condition 1, $h=21.6 \text{ W m}^{-2} \text{ K}^{-1}$ (b) Condition 2, $h=15.9 \text{ W m}^{-2} \text{ K}^{-1}$ **Fig. 7.** Experiments and simulation results of preheating.

The blank area at the lower-right part of the figures means that the heating time is longer than half an hour or the battery cannot be heated to 5 °C at the corresponding condition. Fig. 8 shows that the preheating time decreases with the increase of the alternating current amplitude and the decrease of the frequency. Better thermal insulation conditions can help to reduce the heating time. The maps provide a convenient guide for the selection of the current amplitude and frequency when AC preheating the battery.

It is clear from the above results that AC preheating method with proper selection of amplitude and frequency can achieve good preheating performance and is possible to be used in real preheating devices in electric vehicles to preheat batteries. For example, an alternating current with an amplitude of 7 A and a

frequency of 1 Hz in condition 2 can heat the cell from −20 °C to 5 °C in less than 15 min and the preheating time can be further reduced with the enhancement of thermal insulation.

4.4. Discussions

4.4.1. The effect of AC current amplitude and frequency and thermal insulation conditions on the preheating performance

From Fig. 8, one can see that increasing the current amplitude is more effective when the amplitude is low, for the contour line is sparser at high current amplitude region. At low frequency region, the dominant factor is the current amplitude, while the current frequency begins to affect the heating performance with the increase of the frequency.

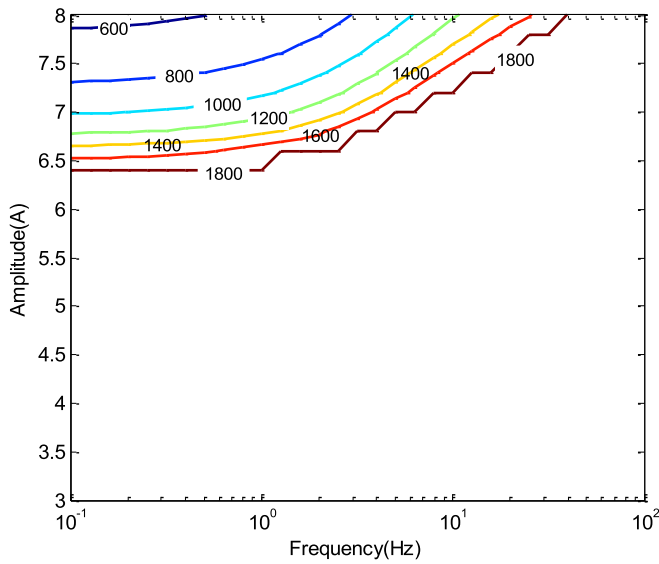
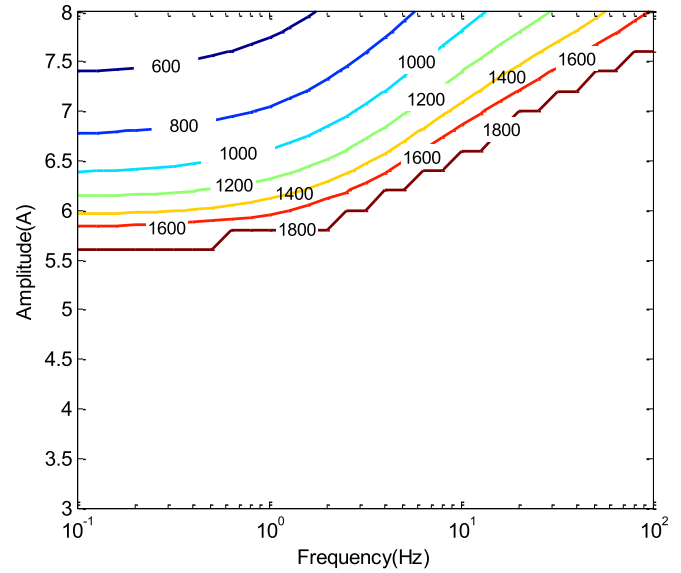
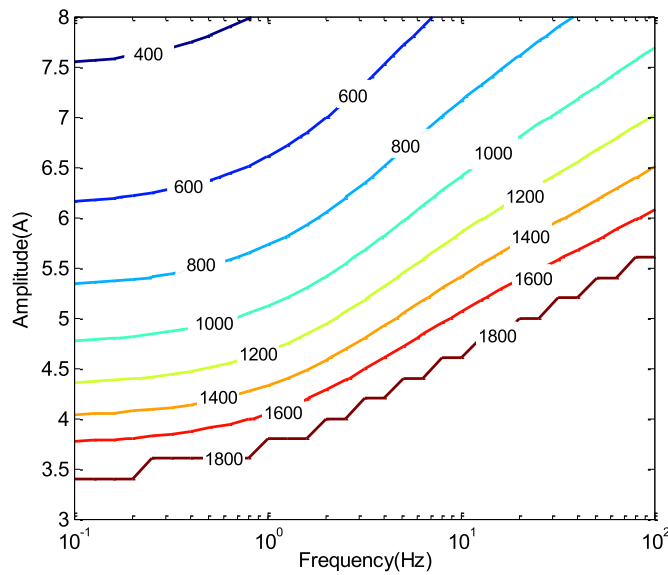
The heating time decreases with the decreasing of the equivalent heat transfer coefficient (h), which indicates that the preheating performance is also very sensitive to the thermal insulation conditions. The relationship between the heating time and the equivalent heat transfer coefficient at 7 A, 1 Hz is displayed in Fig. 9. Results show that the needed preheating time to heat the battery from −20 °C to 5 °C increases by several times with the increasing of equivalent heat transfer coefficient from 0 to 25 $\text{W m}^{-2} \text{ K}^{-1}$.

As mentioned in Section 4.3, the blank area at the lower-right part of Fig. 8 means that the heating time is longer than half an hour or the battery cannot be heated to 5 °C at the corresponding condition. This may be caused by the following reasons: 1) the current amplitude is too low, 2) the current frequency is too high, and 3) the equivalent heat transfer coefficient is too high. For example, the temperature rise at 4 A, 10 Hz with an equivalent heat transfer coefficient of 21.6 $\text{W m}^{-2} \text{ K}^{-1}$ was simulated and plotted in Fig. 10. Under this condition, the battery temperature rising rate is very low and the temperature finally stabilized at about −10 °C where the heat generation and dissipation reach an equilibrium.

4.4.2. The concern on lithium deposition in AC preheating

Besides the consideration of heating performance when selecting AC parameters, the possibility of lithium deposition also needs to be heeded. When AC heating the battery, both the amplitude and the frequency of the current affect the possibility of the occurrence of lithium deposition. In this study, during the AC heating process, a maximum battery voltage around 4.5 V was recorded because of the high current amplitude of ~2 C. This cell voltage should be high enough to cause lithium deposition from the normal consideration of deposition potential. However, the battery capacity was calibrated before and after the repeated preheating experiments. The capacity after the preheating tests remained 2.8 A h and essentially no capacity loss was observed. This is because that the fundamental factors that determine the lithium deposition are the amplitude of the Faradiac current and the magnitude of the integrated value of the Faradiac current within one period (the total charged capacity). The high cell voltage recorded was only instantaneous and larger part of the current was leaked through the non-Faradiac path via the electrical double layer. Much of the high cell voltage was caused by the Ohmic overpotential which will not cause lithium ion deposition.

It is also found in Ref. [12] that there exists a critical charge capacity of 0.1 A h for the 18650 cell used in its study. A charge capacity substantially less than this critical value is unlikely to induce lithium deposition even at −20 °C. In this study, the current frequency is high enough that the charged capacity is well below 0.1 A h in one period. Therefore, the AC heating with the parameters carefully selected is expected to cause no lithium deposition and have no detrimental effect on the battery health.

(a) Condition 1, $h=21.6 \text{ W m}^{-2} \text{ K}^{-1}$ (b) Condition 2, $h=15.9 \text{ W m}^{-2} \text{ K}^{-1}$ (c) Adiabatic condition, $h=0 \text{ W m}^{-2} \text{ K}^{-1}$ **Fig. 8.** Preheating time maps (time unit: s) with different current parameters and thermal insulation conditions.

5. Conclusions

In this study, a method was developed to internally preheat lithium-ion batteries at low temperature using a sinusoidal alternating current. A model in the frequency domain was developed to simulate the heat generation rate based on the EEC. The frequency-domain heat generation model was then adopted in a lumped energy conservation model to explore the relationship between current parameters (amplitudes and frequencies) and preheating effect. The predicted temperature rise was validated against the experimental results and showed good agreement, which indicates that the heat generation model in the

frequency-domain can accurately predict the heat generation rate and the lumped energy conservation model is suitable for the test battery.

The test battery subjected to an alternating current with an amplitude of 7 A (2.25 °C) and a frequency of 1 Hz under an equivalent heat transfer coefficient of $15.9 \text{ W m}^{-2} \text{ K}^{-1}$ can be heated from -20 °C to 5 °C in less than 15 min and the temperature distribution remains essentially uniform. This internal preheating method can be used in real preheating devices of electric vehicles to preheat batteries.

Results in the preheating time maps show that the preheating time decreases with the increase of the alternating current

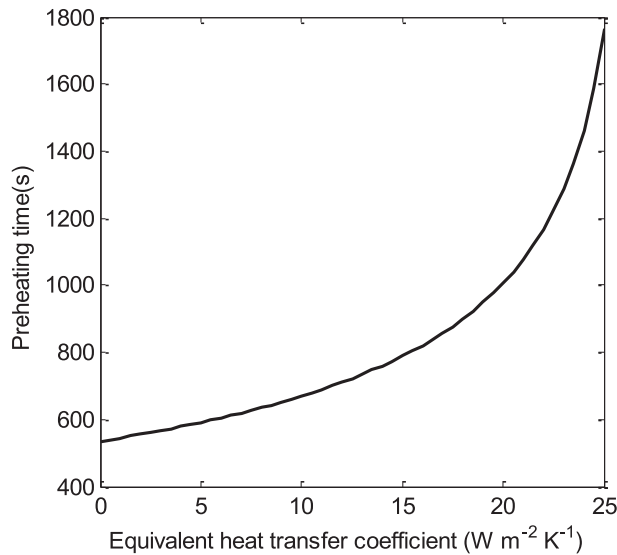


Fig. 9. The relationship between the heating time and the equivalent heat transfer coefficient at 7 A, 1 Hz.

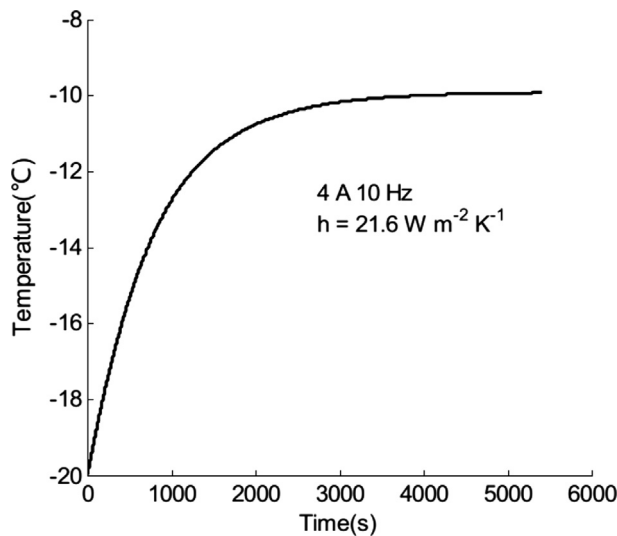


Fig. 10. Temperature rise at 4 A, 10 Hz, $h = 21.6 \text{ W m}^{-2} \text{ K}^{-1}$.

amplitude and the decrease of the frequency. Better thermal insulation conditions can also help to reduce the heating time. Increasing the current amplitude is more effective than adjusting the frequency for improving the heating rate especially in the low-amplitude and low-frequency range. In the higher range of both parameters, the frequency begins to play equivalent role in affecting the heating performance.

Although the test battery experienced instantaneous high voltage during the preheating tests, noticeable capacity loss was not found after the repeated preheating tests. This indicates that the AC preheating method with the suitably selected parameters will not cause lithium deposition and have no detrimental effect on the battery health.

Further study needs to be conducted on how to select the amplitude and the frequency of the AC current so that the internal heating power can be maximized while minimizing the risk of lithium ion deposition.

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References

- [1] T.A. Stuart, A. Hande, J. Power Sources 129 (2004) 368–378.
- [2] C. Alaoui, Z.M. Salameh, IEEE Trans. Veh. Technol. 54 (2005) 468–476.
- [3] A.A. Pesaran, A. Vlahinos, T.A. Stuart, in: The 6th ASME-JSME Thermal Engineering Conference, Hawaii Island, Hawaii, March 17–20, 2003.
- [4] A. Hande, T.A. Stuart, IEEE Power Electron. Transp. (2002) 119–124.
- [5] Hande, in: Proceedings of the 2004 IEEE Workshop on Computers in Power Electronics, 2004, pp. 215–222.
- [6] Y. Ji, C.Y. Wang, Electrochim. Acta 107 (2013) 664–674.
- [7] D. Andre, M. Meiler, K. Steiner, H. Walz, T. Soczka-Guth, D.U. Sauer, J. Power Sources 196 (2011) 5349–5356.
- [8] D. Andre, M. Meiler, K. Steiner, C. Wimmer, T. Soczka-Guth, D.U. Sauer, J. Power Sources 196 (2011) 5334–5341.
- [9] G. Zhang, L. Cao, S. Ge, C. Wang, C.E. Shaffer, C.D. Rahn, in: The Electrochemical Society: Meeting Abstracts, October 2013, p. 538.
- [10] Z. Ogumi, Electrochemistry 78 (2010) 319–324.
- [11] S.J. Drake, D.A. Wetz, J.K. Ostanek, S.P. Miller, J.M. Heinzel, A. Jain, J. Power Sources 252 (2014) 298–304.
- [12] O.J. Kwon, W. Fang, C. Wang, in: The Electrochemical Society: Meeting Abstracts, May 2010, p. 279.